

# MedTech Solutions GmbH

Precision Components for Medical Diagnostics

## Internal Technical Document Collection

RAG Knowledge Base — 10 Documents

|                         |                        |
|-------------------------|------------------------|
| <b>Classification:</b>  | Internal Use Only      |
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| <b>Approved by:</b>     | Dr. Katrin Vogler, CEO |
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# 1 Product Datasheet — MDX-4200-R3 Optical Sensor Array Module

## Product Datasheet: MDX-4200-R3 Optical Sensor Array Module

**Document ID:** DOC-ENG-DS-001  
**Revision:** Rev 3.2  
**Effective Date:** 2026-01-15  
**Department:** Engineering

### 1.1 Product Overview

The **MDX-4200-R3** is a 16-channel optical sensor array module designed for quantitative photometric analysis in clinical blood chemistry analyzers. Revision 3 introduces an improved LED excitation source (PN-OS-4203) with enhanced thermal stability and a redesigned collimating lens assembly (PN-OS-4202) for reduced channel-to-channel crosstalk below  $-45$  dB.

The module integrates with OEM diagnostic platforms via a standardized 40-pin ZIF connector (MTS-CON-40Z) and communicates over SPI at up to 10 MHz.

### 1.2 Key Specifications

Table 1: MDX-4200-R3 Electrical and Optical Specifications

| Parameter                  | Value                       | Conditions            |
|----------------------------|-----------------------------|-----------------------|
| Number of channels         | 16                          | —                     |
| Photodiode type            | Si PIN (InGaAs optional)    | PN-OS-4201            |
| Spectral range             | 340–850 nm                  | —                     |
| Peak responsivity          | $0.45 \pm 0.02$ A/W         | at 450 nm, 25 °C      |
| Dark current (per channel) | $< 2$ nA                    | at $V_R = 5$ V, 25 °C |
| LED excitation wavelength  | $405 \pm 3$ nm              | PN-OS-4203            |
| LED optical power          | 12 mW (typ.)                | at $I_F = 20$ mA      |
| LED lifetime ( $L_{70}$ )  | $> 25,000$ h                | at $T_j = 60$ °C      |
| Filter wheel positions     | 6                           | PN-OS-4204            |
| Filter switching time      | $< 150$ ms                  | position to position  |
| Channel crosstalk          | $< -45$ dB                  | adjacent channels     |
| Supply voltage             | $5.0 \pm 0.25$ V            | main; 3.3 V logic     |
| Power consumption          | $< 1.8$ W                   | all channels active   |
| Operating temperature      | 15–40 °C                    | non-condensing        |
| Dimensions (L×W×H)         | $68 \times 42 \times 18$ mm | excluding connector   |
| Weight                     | 54 g                        | —                     |

### 1.3 Bill of Materials (Key Components)

Table 2: MDX-4200-R3 Key Component BOM

| Part Number | Description                     | Qty | Supplier                                  |
|-------------|---------------------------------|-----|-------------------------------------------|
| PN-OS-4201  | 16-ch Si PIN photodiode array   | 1   | SUP-001 (Hamamatsu)                       |
| PN-OS-4202  | Collimating lens assembly (BK7) | 1   | SUP-006 (Schott AG)                       |
| PN-OS-4203  | LED excitation source, 405 nm   | 1   | SUP-001 (Hamamatsu)                       |
| PN-OS-4204  | 6-position optical filter wheel | 1   | In-house ( <a href="#">PROC-ENG-101</a> ) |
| PN-SP-5501  | 24-bit ADC, 1 kSPS              | 2   | SUP-003 (Texas Instruments)               |
| PN-SP-5502  | Transimpedance amplifier IC     | 16  | SUP-003 (Texas Instruments)               |
| PN-SP-5503  | EMI shielding enclosure         | 1   | SUP-007 (Würth Elektronik)                |

### 1.4 Calibration Requirements

The MDX-4200-R3 requires photometric calibration per [PROC-CAL-401](#) using NIST-traceable reference standards. Calibration interval: **every 2,000 operating hours or 12 months**, whichever comes first. Calibration data is stored in the on-board EEPROM (256 kB) and can be read via SPI command 0xA3.

### 1.5 Relevant Failure Codes

- **FC-OPT-01**: Photodiode dark current out of spec ( $> 2 \text{ nA}$ )
- **FC-OPT-02**: LED intensity degradation ( $> 15\%$  from nominal)
- **FC-OPT-03**: Filter wheel indexing error
- **FC-ELC-01**: ADC linearity deviation ( $> \pm 3 \text{ LSB}$ )
- **FC-ELC-02**: Power supply ripple ( $> 10 \text{ mV}_{\text{pp}}$  on 3.3 V rail)

### 1.6 Applicable Standards

- IEC 61010-1:2010+A1:2019 (Safety)
- EN 61326-1:2013 (EMC — Emission and Immunity)
- ISO 13485:2016 (Quality Management System)
- EU IVDR 2017/746 (In Vitro Diagnostic Regulation)

### 1.7 Revision History

| Rev | Date       | Author          | Changes                        |
|-----|------------|-----------------|--------------------------------|
| 3.0 | 2025-06-01 | Dr. L. Pfeiffer | Initial R3 release             |
| 3.1 | 2025-09-20 | M. Brenner      | Updated LED lifetime data      |
| 3.2 | 2026-01-15 | Dr. L. Pfeiffer | Added filter wheel timing spec |

## 2 Standard Operating Procedure — Cleanroom Assembly of MDX-4200

### SOP: Cleanroom Assembly of MDX-4200-R3

**Document ID:** DOC-PROD-SOP-002  
**Revision:** Rev 2.1  
**Effective Date:** 2025-11-10  
**Department:** Production

#### 2.1 Purpose

This SOP defines the step-by-step assembly procedure for the MDX-4200-R3 Optical Sensor Array Module inside the MedTech Solutions ISO 7 cleanroom (Room CR-A12, Building 2). It covers handling of sensitive optical and electronic components, bonding sequences, and in-process verification checkpoints.

#### 2.2 Scope

Applicable to all production technicians and engineers involved in **PROC-PROD-301** (Cleanroom Assembly) and **PROC-ENG-101** (Optical Alignment and Bonding).

#### 2.3 Prerequisites

- Operator must hold valid cleanroom certification (MTS-CERT-CR, renewed annually).
- All incoming components must have passed **PROC-QA-201** (Incoming Goods Inspection) with status **QA-PASS**.
- Cleanroom particle count verified  $\leq 352,000$  particles/m<sup>3</sup> ( $\geq 0.5\ \mu\text{m}$ ) per ISO 14644-1 Class 7.
- ESD wrist strap resistance verified  $< 35\ \text{M}\Omega$ .

#### 2.4 Required Materials and Tools

Table 3: Materials and Tools for MDX-4200-R3 Assembly

| Item               | Part / Tool ID                           | Notes                                     |
|--------------------|------------------------------------------|-------------------------------------------|
| Photodiode array   | PN-OS-4201                               | ESD-sensitive, handle with vacuum pen     |
| Collimating lens   | PN-OS-4202                               | Store in N <sub>2</sub> cabinet until use |
| LED source         | PN-OS-4203                               | Verify forward voltage before mounting    |
| Filter wheel       | PN-OS-4204                               | Pre-assembled, verify indexing            |
| Signal PCB         | (sub-assembly of PN-SP-5501, PN-SP-5502) | From <b>PROC-ENG-102</b>                  |
| UV-curing adhesive | MTS-ADH-UV-07                            | Lot number must be recorded               |
| Alignment fixture  | MTS-JIG-OA-42                            | Calibrated quarterly                      |
| UV curing lamp     | MTS-UV-LAMP-01                           | 365 nm, 80 mW/cm <sup>2</sup>             |
| Torque driver      | MTS-TOOL-TQ-03                           | Set to 0.15 Nm for M2 screws              |

## 2.5 Assembly Procedure

- Step 1: PCB Preparation.** Retrieve the populated signal processing PCB from the SMT line ([PROC-ENG-102](#)). Perform visual inspection under  $10\times$  magnification. Verify solder joint quality per IPC-A-610 Class 3. Record PCB serial number in MES (Manufacturing Execution System) under batch ID.
- Step 2: EMI Shield Mounting.** Place the EMI shielding enclosure (PN-SP-5503) onto the PCB. Secure with four M2 $\times$ 4 screws at  $0.15 \pm 0.02$  Nm. Verify continuity between shield and PCB ground plane ( $< 50$  m $\Omega$ ).
- Step 3: Photodiode Array Installation.** Using vacuum pen (tip size 3), place PN-OS-4201 onto the optical bench alignment pins. Apply  $2.0 \pm 0.3$   $\mu$ L of UV-curing adhesive (MTS-ADH-UV-07) to the four corner pads. **Do not cure yet.**
- Step 4: Collimating Lens Mounting.** Insert PN-OS-4202 into the lens holder. Ensure the flat reference edge is oriented toward fiducial mark FM-A on the housing. Apply adhesive to the retaining ring.
- Step 5: Optical Alignment.** Mount the sub-assembly into alignment fixture MTS-JIG-OA-42. Activate the alignment laser (635 nm reference). Adjust X/Y micrometers until all 16 channels show centroid deviation  $< 5$   $\mu$ m on the alignment camera. Record final positions in the MES.
- Step 6: UV Curing.** Expose adhesive joints to UV lamp MTS-UV-LAMP-01 for  $60 \pm 5$  seconds at 80 mW/cm<sup>2</sup>. Verify cure by checking adhesive hardness (Shore D  $> 75$ ).
- Step 7: LED Source Installation.** Solder PN-OS-4203 to the designated pads on the PCB. Verify forward voltage  $V_F = 3.4 \pm 0.2$  V at  $I_F = 20$  mA. Record measured  $V_F$  in the device history record (DHR).
- Step 8: Filter Wheel Integration.** Attach the pre-assembled filter wheel (PN-OS-4204) to the stepper motor shaft. Run the self-test routine (`CMD_FWHEEL_HOME`) and verify that all 6 positions are reached within  $\pm 0.3^\circ$  of nominal. If [FC-OPT-03](#) is triggered, return to Step 5 for re-alignment.
- Step 9: Enclosure and Potting.** Proceed to [PROC-PROD-302](#) (Potting and Encapsulation). Apply conformal coating to the PCB per drawing MTS-DWG-4200-05. Close the housing and secure with six M2.5 $\times$ 6 screws at  $0.20 \pm 0.02$  Nm.
- Step 10: Laser Marking.** Transfer to [PROC-PROD-303](#). Mark the unit with serial number, production date (YYYY-WW format), and 2D data matrix code. Verify readability with barcode scanner (Grade C minimum per ISO/IEC 15415).
- Step 11: In-Process Verification.** Execute [PROC-QA-202](#) (Optical Verification). All 16 channels must read within  $\pm 3\%$  of the reference standard. Units failing this step are tagged with [FC-OPT-01](#) or [FC-OPT-02](#) and routed to the deviation handling process (see DOC-QA-QR-004, Section 4).

## 2.6 Records

All assembly data, measurements, and deviations must be recorded in the MES (System: MTS-MES-Pro v4.2) under the corresponding production order number. Device History Records are archived for a minimum of 15 years per ISO 13485 and EU IVDR requirements.



### 3 Troubleshooting Guide — MDX-4200 Optical Sensor Array Module

#### Troubleshooting Guide: MDX-4200 Series

**Document ID:** DOC-CS-TSG-003  
**Revision:** Rev 1.4  
**Effective Date:** 2025-12-05  
**Department:** Customer Support / Engineering

#### 3.1 Purpose

This guide provides systematic troubleshooting procedures for field issues reported on the MDX-4200 series (all revisions). It is intended for MedTech Solutions field service engineers and trained OEM technical staff.

#### 3.2 Diagnostic Access

Connect to the MDX-4200 diagnostic interface via the 40-pin ZIF connector using the MTS-DIAG-KIT-01 evaluation board. Launch the **MTS Diagnostics Suite v2.8** software on a Windows 10/11 PC. The module is addressed via SPI; use the `/scan` command to detect connected units.

#### 3.3 Symptom–Cause–Action Table

| Symptom                                                                   | Probable Cause                                                      | Failure Code     | Corrective Action                                                                                                                                                                                                     |
|---------------------------------------------------------------------------|---------------------------------------------------------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| One or more channels show elevated baseline signal (> 500 counts at dark) | Photodiode dark current exceeds 2 nA due to thermal stress or aging | <b>FC-OPT-01</b> | Run dark-current test via <b>CMD_DARK_TEST</b> . If > 3 channels affected, replace photodiode array PN-OS-4201.<br>Single-channel failures may indicate a solder joint defect — re-flow affected pads.                |
| Measured absorbance values consistently 15–30% lower than expected        | LED intensity degradation; optical path contamination               | <b>FC-OPT-02</b> | Run <b>CMD_LED_ITEST</b> to measure relative LED power. If < 85% of factory value, replace LED PN-OS-4203. Also clean collimating lens PN-OS-4202 with IPA and lint-free wipe. Re-calibrate per <b>PROC-CAL-401</b> . |

*Continued on next page*

| Symptom                                               | Probable Cause                                                           | Failure Code | Corrective Action                                                                                                                                                                                                                        |
|-------------------------------------------------------|--------------------------------------------------------------------------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Filter wheel does not home or skips positions         | Stepper motor driver fault; mechanical obstruction; encoder misalignment | FC-OPT-03    | Power-cycle the module. Run <code>CMD_FWHEEL_HOME</code> . If error persists, check motor connector (J4) seating. Inspect filter wheel PN-0S-4204 for debris. If encoder index pulse is missing, escalate to Level 2 (RMA required).     |
| Noisy or drifting ADC readings on all channels        | Power supply ripple on 3.3 V rail; grounding issue                       | FC-ELC-02    | Measure 3.3 V rail ripple with oscilloscope (BW $\geq 200$ MHz). If $> 10$ mV <sub>pp</sub> , check decoupling capacitors C12, C15, C18 on the signal PCB. Verify EMI shield PN-SP-5503 grounding continuity.                            |
| ADC output shows non-linear response curve            | ADC calibration drift or IC defect                                       | FC-ELC-01    | Run <code>CMD_ADC_LINTTEST</code> with calibration source. If deviation $> \pm 3$ LSB, replace ADC PN-SP-5501. Requires re-calibration per PROC-CAL-401.                                                                                 |
| Module fails EMC pre-compliance test at customer site | EMI shielding integrity compromised                                      | FC-ELC-03    | Verify all shield screws are torqued to $0.15 \pm 0.02$ Nm. Check gasket condition on EMI enclosure PN-SP-5503. If shield was opened for service, ensure conductive gasket (MTS-GSKT-EMI-01) is properly seated. Re-test per EN 61326-1. |
| Serial number or 2D code unreadable                   | Laser marking degradation                                                | FC-MEC-02    | Re-mark unit per PROC-PROD-303. If housing surface is damaged, replace housing (MTS-HSG-4200-01).                                                                                                                                        |

### 3.4 Escalation Procedure

- Level 1** (Field Service): Follow symptom–cause–action table above.
- Level 2** (Engineering Support): Contact `eng-support@medtech-solutions.de` with diagnostic log file (`.mtslog`) and photos.
- Level 3** (RMA): If Level 2 cannot resolve, initiate RMA via the MTS Customer Portal. Reference the failure code and incident number. Typical RMA turnaround: 10 business days.

## 4 Quarterly Quality Report — Q1 2026

### Quarterly Quality Report: Q1 2026 (January–March)

**Document ID:** DOC-QA-QR-004  
**Revision:** Rev 1.0  
**Effective Date:** 2026-04-02  
**Department:** Quality Assurance

#### 4.1 Executive Summary

This report summarizes quality performance across all product lines for Q1 2026. Overall first-pass yield improved to **96.8%** (up from 95.3% in Q4 2025), driven by the corrective action on solder paste stencil design for the MDX-5500 signal processing PCB (see CAPA-2025-041). Customer complaint rate decreased to **0.34%** (target: < 0.50%).

#### 4.2 Production Volume and Yield

Table 5: Q1 2026 Production Summary by Product Line

| Product      | Units Produced | Units Passed | FPY (%)     | Trend           |
|--------------|----------------|--------------|-------------|-----------------|
| MDX-4200-R3  | 1,240          | 1,198        | 96.6        | ↑ (+0.8)        |
| MDX-3100     | 860            | 841          | 97.8        | ↑ (+1.2)        |
| MDX-5500     | 2,100          | 2,031        | 96.7        | ↑ (+2.1)        |
| MDX-6100     | 480            | 464          | 96.7        | ↔ (0.0)         |
| MDX-7000     | 5,600          | 5,404        | 96.5        | ↓ (−0.3)        |
| <b>Total</b> | <b>10,280</b>  | <b>9,938</b> | <b>96.8</b> | <b>↑ (+1.5)</b> |

#### 4.3 Top Failure Modes

Table 6: Top 5 Failure Codes in Q1 2026

| Rank | Failure Code | Occurrences | Rate (%) | Affected Product |
|------|--------------|-------------|----------|------------------|
| 1    | FC-FLD-03    | 87          | 1.55     | MDX-7000         |
| 2    | FC-OPT-02    | 24          | 0.23     | MDX-4200-R3      |
| 3    | FC-ELC-02    | 18          | 0.18     | MDX-5500         |
| 4    | FC-THR-02    | 12          | 0.12     | MDX-6100         |
| 5    | FC-MEC-01    | 9           | 0.09     | MDX-7000         |

#### 4.4 Failure Analysis: FC-FLD-03 (Reagent Blister Rupture)

The highest-volume failure in Q1 was **FC-FLD-03**, affecting 87 units of the MDX-7000 diagnostic cartridge. Root cause analysis identified a change in foil thickness by the blister film supplier (lot BL-2025-1187 from SUP-005, microfluidic ChipShop GmbH). The foil was measured at  $48 \pm 4 \mu\text{m}$  versus the specified  $55 \pm 3 \mu\text{m}$ , reducing burst strength below the minimum threshold of 180 kPa.

**Corrective Action (CAPA-2026-008):**

- Immediate: Quarantine of affected lot; 100% incoming inspection of foil thickness per **PROC-QA-201**.
- Short-term: Supplier audit of SUP-005 scheduled for 2026-04-15.
- Long-term: Dual-source qualification for blister film (SUP-008, Zeon Corporation, as backup).

#### 4.5 Supplier Quality Summary

Table 7: Supplier Quality Performance Q1 2026

| Supplier ID | Supplier Name         | Lots Received | Lots Rejected | Rating           |
|-------------|-----------------------|---------------|---------------|------------------|
| SUP-001     | Hamamatsu Photonics   | 14            | 0             | A (Excellent)    |
| SUP-002     | Bürkert Fluid Control | 8             | 0             | A (Excellent)    |
| SUP-003     | Texas Instruments     | 22            | 1             | B (Good)         |
| SUP-004     | Laird Thermal Systems | 6             | 0             | A (Excellent)    |
| SUP-005     | microfluidic ChipShop | 18            | 3             | C (Under Review) |
| SUP-006     | Schott AG             | 10            | 0             | A (Excellent)    |
| SUP-007     | Würth Elektronik      | 12            | 0             | A (Excellent)    |
| SUP-008     | Zeon Corporation      | 4             | 0             | A (Excellent)    |

#### 4.6 Customer Complaints

Table 8: Customer Complaints Q1 2026

| Complaint ID | Product     | Issue                                          | Status                  |
|--------------|-------------|------------------------------------------------|-------------------------|
| CC-2026-001  | MDX-4200-R3 | Filter wheel intermittent ( <b>FC-OPT-03</b> ) | Closed (RMA)            |
| CC-2026-002  | MDX-7000    | Cartridge leakage during transport             | Open (CAPA-2026-008)    |
| CC-2026-003  | MDX-6100    | Thermal overshoot during first 5 min           | Closed (FW update v2.4) |
| CC-2026-004  | MDX-5500    | EMC test marginal at 150 MHz                   | Under investigation     |

#### 4.7 Management Review Actions

1. Complete SUP-005 audit by 2026-04-30 (Owner: Dr. S. Hartmann).
2. Validate dual-source for blister film by Q2 2026 (Owner: J. Krickl).
3. Investigate EMC margin on MDX-5500 at 150 MHz (Owner: Dr. L. Pfeiffer).
4. Review MDX-7000 housing snap-fit design to reduce **FC-MEC-01** (Owner: M. Brenner).

## 5 Supplier Component Specification — Photodiode Array (SUP-001)

### Supplier Component Specification: 16-Channel Si PIN Photodiode Array

**Document ID:** DOC-ENG-SS-005  
**Revision:** Rev 2.0  
**Effective Date:** 2025-08-22  
**Department:** Engineering / Procurement

#### 5.1 Scope

This specification defines the requirements for the 16-channel silicon PIN photodiode array procured from **SUP-001 (Hamamatsu Photonics K.K., Hamamatsu, Japan)** for use in the MDX-4200 Optical Sensor Array Module. The component is assigned internal part number PN-OS-4201.

#### 5.2 Supplier Information

| Field                | Value                                                                                  |
|----------------------|----------------------------------------------------------------------------------------|
| Supplier ID          | SUP-001                                                                                |
| Company              | Hamamatsu Photonics K.K.                                                               |
| Address              | 325-6 Sunayama-cho, Naka-ku, Hamamatsu 430-8587, Japan                                 |
| Contact (Technical)  | Mr. Takeshi Yamada, <a href="mailto:t.yamada@hamamatsu.com">t.yamada@hamamatsu.com</a> |
| Contact (Commercial) | Ms. Yuki Tanaka, <a href="mailto:y.tanaka@hamamatsu.com">y.tanaka@hamamatsu.com</a>    |
| Hamamatsu Part No.   | S11639-01 (custom variant for MTS)                                                     |
| MTS Part No.         | PN-OS-4201                                                                             |
| Quality Agreement    | QA-SUP001-2023-v2 (signed 2023-03-15)                                                  |

### 5.3 Technical Requirements

Table 9: PN-OS-4201 Performance Requirements

| Parameter                                        | Min                  | Typ                 | Max                 |
|--------------------------------------------------|----------------------|---------------------|---------------------|
| Number of elements                               |                      | 16                  |                     |
| Active area per element                          |                      | $1.2 \times 2.4$ mm |                     |
| Element pitch                                    |                      | 2.5 mm              |                     |
| Spectral response range                          | 320 nm               | —                   | 900 nm              |
| Peak wavelength                                  | —                    | 450 nm              | —                   |
| Responsivity @ 450 nm                            | 0.43 A/W             | 0.45 A/W            | 0.47 A/W            |
| Dark current ( $V_R = 5$ V, $25^\circ\text{C}$ ) | —                    | 0.5 nA              | 2.0 nA              |
| Rise time ( $V_R = 5$ V, $R_L = 1$ k $\Omega$ )  | —                    | 3.0 $\mu\text{s}$   | 5.0 $\mu\text{s}$   |
| Terminal capacitance ( $V_R = 5$ V)              | —                    | 15 pF               | 25 pF               |
| Crosstalk (adjacent channels)                    | —                    | −50 dB              | −45 dB              |
| Operating temperature                            | −10 $^\circ\text{C}$ | —                   | 60 $^\circ\text{C}$ |
| Storage temperature                              | −20 $^\circ\text{C}$ | —                   | 80 $^\circ\text{C}$ |

### 5.4 Quality and Inspection Requirements

- Acceptance Quality Limit (AQL): 0.65% for critical parameters (responsivity, dark current); 1.5% for non-critical.
- Each lot must include a Certificate of Conformance (CoC) with measured values for all parameters in the specification table.
- MedTech Solutions reserves the right to perform incoming inspection per [PROC-QA-201](#) on a sampling basis (General Inspection Level II, single sampling, normal).
- Supplier must notify MTS of any process change, material change, or manufacturing site change at least 90 days in advance (per QA-SUP001-2023-v2, Section 7.3).

### 5.5 Packaging and Labeling

- Individual trays (50 pcs per tray) in ESD-safe packaging.
- Each tray labeled with: Hamamatsu part number, lot number, quantity, date of manufacture, and MTS PO number.
- Moisture Sensitivity Level (MSL): MSL 3 per IPC/JEDEC J-STD-020. Floor life after opening: 168 hours at  $\leq 30^\circ\text{C}$  / 60% RH.

### 5.6 Lead Time and MOQ

| Parameter                     | Value                          |
|-------------------------------|--------------------------------|
| Standard lead time            | 12 weeks ARO                   |
| Minimum order quantity        | 200 pcs                        |
| Price (at 1,000 pcs/year)     | €18.40 per unit (CIF Freiburg) |
| Annual volume forecast (2026) | 1,600 pcs                      |

## 6 Maintenance Log — Production Equipment (CR-A12)

### Maintenance Log: Cleanroom CR-A12 Production Equipment

**Document ID:** DOC-PROD-ML-006  
**Revision:** Rev 3.0  
**Effective Date:** 2026-03-31  
**Department:** Production / Facilities

#### 6.1 Purpose

This log records all preventive and corrective maintenance activities for production equipment located in Cleanroom CR-A12, Building 2. Equipment maintenance is critical for sustaining product quality and is audited annually under ISO 13485.

#### 6.2 Equipment Register

Table 10: CR-A12 Equipment Register

| Equipment ID | Description                               | Manufacturer         | PM Interval |
|--------------|-------------------------------------------|----------------------|-------------|
| EQ-CR12-001  | SMT Reflow Oven (10-zone)                 | Rehm Thermal Systems | Monthly     |
| EQ-CR12-002  | Pick-and-Place Machine                    | ASM Assembly Systems | Bi-weekly   |
| EQ-CR12-003  | UV Curing Station (MTS-UV-LAMP-01)        | Dymax Corporation    | Quarterly   |
| EQ-CR12-004  | Optical Alignment Station (MTS-JIG-OA-42) | In-house             | Quarterly   |
| EQ-CR12-005  | Solder Paste Printer                      | DEK / ASM            | Monthly     |
| EQ-CR12-006  | AOI System (Automated Optical Insp.)      | Koh Young Technology | Semi-annual |
| EQ-CR12-007  | Wire Bonding Machine                      | Kulicke & Soffa      | Quarterly   |
| EQ-CR12-008  | Laser Marking System                      | FOBA (ALLTEC)        | Semi-annual |

#### 6.3 Maintenance Records — Q1 2026

| Date       | Equip. ID   | Type | Description                                                                                                                                        | Technician |
|------------|-------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| 2026-01-06 | EQ-CR12-001 | PM   | Monthly profile verification: all 10 zones within $\pm 2^{\circ}\text{C}$ of set-point. Replaced conveyor belt (wear detected at Zone 7 entry).    | T. Müller  |
| 2026-01-08 | EQ-CR12-002 | PM   | Bi-weekly nozzle cleaning and feeder calibration. Replaced nozzle N-08 (vacuum leak). Placement accuracy verified at $\pm 25\text{ }\mu\text{m}$ . | A. Weber   |
| 2026-01-14 | EQ-CR12-005 | PM   | Monthly squeegee replacement and stencil tension check. Tension: $42\text{ N/cm}$ (spec: $40 \pm 5\text{ N/cm}$ ).                                 | T. Müller  |

| Date       | Equip. ID   | Type | Description                                                                                                                                                                                                                                             | Technician        |
|------------|-------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| 2026-01-22 | EQ-CR12-002 | PM   | Bi-weekly check. All feeders OK. Vision alignment camera cleaned.                                                                                                                                                                                       | A. Weber          |
| 2026-02-03 | EQ-CR12-001 | PM   | Monthly profile check. Zone 4 1.8 °C low — adjusted heater PID.                                                                                                                                                                                         | T. Müller         |
| 2026-02-05 | EQ-CR12-002 | PM   | Bi-weekly check. Replaced feeder F-12 (intermittent misfeed on 0402 parts).                                                                                                                                                                             | A. Weber          |
| 2026-02-10 | EQ-CR12-003 | PM   | Quarterly UV intensity measurement: 78 mW/cm <sup>2</sup> (spec: ≥ 75). Bulb nearing replacement threshold. Scheduled for Q2.                                                                                                                           | K. Fischer        |
| 2026-02-10 | EQ-CR12-004 | PM   | Quarterly calibration of alignment fixture MTS-JIG-OA-42. X/Y repeatability: ±1.8 μm (spec: ±3 μm). Laser reference aligned.                                                                                                                            | Dr. L. Pfeiffer   |
| 2026-02-14 | EQ-CR12-005 | PM   | Monthly maintenance. Stencil replaced (lot change to MDX-5500 Rev B layout).                                                                                                                                                                            | T. Müller         |
| 2026-02-19 | EQ-CR12-002 | CM   | <b>Corrective:</b> Machine halted — head 2 encoder fault. Encoder cable replaced. Downtime: 3.5 h. Impact: 45 units delayed (MDX-4200-R3 batch PO-2026-0218).                                                                                           | A. Weber          |
| 2026-03-02 | EQ-CR12-001 | PM   | Monthly check. All zones nominal. Flux trap cleaned.                                                                                                                                                                                                    | T. Müller         |
| 2026-03-04 | EQ-CR12-006 | PM   | Semi-annual full calibration. Reference board scanned: all defect types detected. Software updated to v6.1.2.                                                                                                                                           | Koh Young Service |
| 2026-03-10 | EQ-CR12-008 | PM   | Semi-annual laser power check: 18.2 W (spec: 18 ± 2 W). Lens cleaned. Marking samples verified readable (Grade B).                                                                                                                                      | K. Fischer        |
| 2026-03-15 | EQ-CR12-007 | PM   | Quarterly bond pull test: average 8.2 gf (spec: ≥ 6 gf). Capillary replaced (32,000 bonds).                                                                                                                                                             | A. Weber          |
| 2026-03-20 | EQ-CR12-004 | CM   | <b>Corrective:</b> Alignment laser drift detected during MDX-4200-R3 production (batch PO-2026-0312). 12 units quarantined for re-verification. Laser module replaced. Root cause: thermal expansion of mounting bracket. Added thermal insulation pad. | Dr. L. Pfeiffer   |



## 6.4 Open Maintenance Actions

- **EQ-CR12-003:** UV bulb replacement scheduled for April 2026 (approaching 2,000 h limit).
- **EQ-CR12-004:** Thermal insulation effectiveness to be verified over next 3 production batches.
- **EQ-CR12-002:** Head 2 encoder monitoring — daily check added until end of April.

## 7 ISO 13485:2016 Compliance Checklist — Annual Internal Audit 2026

### ISO 13485:2016 Compliance Checklist

**Document ID:** DOC-QA-CC-007  
**Revision:** Rev 1.0  
**Effective Date:** 2026-03-01  
**Department:** Quality Assurance

#### 7.1 Audit Information

| Field               | Value                                                |
|---------------------|------------------------------------------------------|
| Audit type          | Internal surveillance audit (annual)                 |
| Audit dates         | 2026-03-03 to 2026-03-05                             |
| Lead auditor        | Dr. Simone Hartmann (QA Director)                    |
| Co-auditor          | Thomas Klein (External, TÜV SÜD)                     |
| Departments audited | Engineering, Production, QA, Customer Support        |
| Previous audit      | 2025-03-06 to 2025-03-08 (0 major, 2 minor findings) |

#### 7.2 Checklist Summary

| Clause | Requirement                         | Status    | Evidence / Notes                                                                            |
|--------|-------------------------------------|-----------|---------------------------------------------------------------------------------------------|
| 4.1    | QMS scope and processes defined     | Compliant | QMS Manual MTS-QM-001 Rev 8. Process map updated Jan 2026.                                  |
| 4.2.4  | Document control                    | Compliant | All documents controlled via MTS-DMS v3.1. Reviewed 50 documents; all current.              |
| 4.2.5  | Record control                      | Compliant | DHRs archived in secure server (15-year retention). Spot-checked 20 records — all complete. |
| 5.6    | Management review                   | Compliant | Last review: 2026-02-15. Minutes documented in MR-2026-Q1.                                  |
| 6.2    | Competence and training             | Compliant | Training matrix current. Cleanroom cert (MTS-CERT-CR) verified for all CR-A12 operators.    |
| 6.3    | Infrastructure and work environment | Compliant | Cleanroom CR-A12 particle counts logged daily. Equipment maintenance per DOC-PROD-ML-006.   |
| 6.4    | Work environment                    | Compliant | Temperature: 21±1 °C, RH: 45±10%. Monitoring via BMS.                                       |

| Clause | Requirement                        | Status        | Evidence / Notes                                                                                         |
|--------|------------------------------------|---------------|----------------------------------------------------------------------------------------------------------|
| 7.1    | Product realization planning       | Compliant     | Design plans for MDX-4200-R3 and MDX-7000 reviewed.                                                      |
| 7.3    | Design and development             | Compliant     | Design history files complete for all active products. V&V protocols executed.                           |
| 7.4.1  | Purchasing process                 | Minor Finding | Supplier evaluation for SUP-005 overdue (last eval: 2024-11-10, interval: 12 months). See CAPA-2026-008. |
| 7.4.3  | Verification of purchased product  | Compliant     | PROC-QA-201 executed for all incoming lots in Q1. Records complete.                                      |
| 7.5.1  | Production and service provision   | Compliant     | SOPs current (DOC-PROD-SOP-002 verified). Process validations up to date.                                |
| 7.5.2  | Cleanliness of product             | Compliant     | Cleaning validation for MDX-7000 cartridge (VR-CLN-7000-v2) current.                                     |
| 7.5.3  | Installation and servicing         | N/A           | OEM responsibility per distribution agreements.                                                          |
| 7.5.9  | Traceability                       | Compliant     | Full traceability via MES serial numbers. Tested forward/backward trace on 5 units.                      |
| 7.6    | Monitoring and measuring equipment | Compliant     | Calibration records current for all equipment in CR-A12 (see DOC-PROD-ML-006).                           |
| 8.2.1  | Feedback and complaints            | Compliant     | Complaint handling per SOP-QA-COMPL-01. 4 complaints in Q1 (see DOC-QA-QR-004).                          |
| 8.2.4  | Product monitoring                 | Compliant     | Final functional test PROC-QA-203 executed on 100% of units.                                             |
| 8.3    | Nonconforming product              | Compliant     | Nonconforming product quarantine area labeled. Disposition records reviewed.                             |
| 8.5.2  | Corrective action                  | Minor Finding | CAPA-2025-039 (from previous audit): effectiveness check was 15 days late. Process otherwise robust.     |
| 8.5.3  | Preventive action                  | Compliant     | Risk register updated Q4 2025. Trend analysis per DOC-QA-QR-004.                                         |

### 7.3 Audit Findings Summary

| Finding                                          | Clause | Due Date for CAPA |
|--------------------------------------------------|--------|-------------------|
| Minor #1: SUP-005 evaluation overdue             | 7.4.1  | 2026-04-30        |
| Minor #2: CAPA-2025-039 effectiveness check late | 8.5.2  | 2026-04-15        |

No major non-conformities identified. Audit conclusion: QMS continues to meet ISO 13485:2016 requirements. Next external surveillance audit by TÜV SÜD scheduled for September 2026.

## 8 Customer FAQ — MDX-7000 Integrated Diagnostic Cartridge

### Customer FAQ: MDX-7000 Integrated Diagnostic Cartridge

**Document ID:** DOC-CS-FAQ-008  
**Revision:** Rev 2.3  
**Effective Date:** 2026-02-20  
**Department:** Customer Support

#### 8.1 General Questions

##### Q1: What is the MDX-7000?

The MDX-7000 is a single-use, point-of-care diagnostic cartridge manufactured by MedTech Solutions GmbH. It integrates microfluidic sample handling, reagent storage, and optical detection into a compact, injection-molded cartridge. It is designed for use with compatible OEM analyzer platforms.

##### Q2: What sample types does the MDX-7000 support?

The cartridge accepts capillary whole blood (10–50  $\mu\text{L}$ ) via the integrated capillary blood inlet port (PN-DC-7003). Venous blood samples can be used with the optional adapter (MTS-ADP-VEN-01). Serum and plasma are not validated.

##### Q3: What analytes can the MDX-7000 measure?

The standard configuration supports: glucose, HbA1c, total cholesterol, HDL, LDL (calculated), triglycerides, and CRP. Custom analyte panels are available upon request (minimum order: 10,000 cartridges).

##### Q4: What is the shelf life of the MDX-7000?

18 months from the date of manufacture when stored at 2–30 °C and < 70% RH. The reagent blister pack (PN-DC-7002) contains lyophilized reagents that are sensitive to moisture. Do not open the foil pouch until immediately before use.

##### Q5: How should the MDX-7000 be stored?

Store in the original foil pouch at 2–30 °C, away from direct sunlight. Do not freeze. Cartridges stored above 35 °C for more than 48 hours must be discarded. Check the expiration date on the pouch label before use.

#### 8.2 Technical Questions

##### Q6: What is the sample volume accuracy?

The microfluidic chip (PN-DC-7001, injection-molded cyclic olefin copolymer) delivers a metered volume of  $25 \pm 0.5 \mu\text{L}$  to the reaction chamber. Overall fluidic volume accuracy is  $\pm 2\%$  for volumes > 10  $\mu\text{L}$ , validated per [PROC-CAL-403](#).

##### Q7: What is the time to result?

Typical time to result is 8–12 minutes depending on the analyte panel. CRP results are available within 5 minutes; HbA1c requires the full 12 minutes due to the enzymatic reaction kinetics.

##### Q8: How does the MDX-7000 interface with the analyzer?

The cartridge is inserted into the analyzer's cartridge bay. Electrical contact is made through 12 gold-plated spring pins on the cartridge base. The analyzer reads the cartridge ID via a 2D data matrix code (laser-marked per [PROC-PROD-303](#)) and automatically selects the appropriate assay protocol.

##### Q9: Can the MDX-7000 be re-used?

No. The MDX-7000 is a single-use device. Attempting to re-use the cartridge will result in

inaccurate results and may damage the analyzer. Each cartridge contains a one-time-use cryptographic token that is invalidated after the first run.

### 8.3 Troubleshooting

#### Q10: The analyzer displays “Cartridge Read Error.” What should I do?

1. Remove the cartridge and inspect the 2D data matrix code for damage. If unreadable (see **FC-MEC-02**), discard and use a new cartridge.
2. Check the gold contact pads on the cartridge base for contamination. Clean gently with a dry lint-free wipe.
3. Verify that the cartridge is not expired.
4. If the problem persists, contact MedTech Solutions support at [support@medtech-solutions.de](mailto:support@medtech-solutions.de) or call +49 761 555 8410.

#### Q11: Results show “QC Fail — Reagent Error.” What does this mean?

This indicates that the internal quality control check detected a reagent integrity issue, likely related to the reagent blister pack (PN-DC-7002). Possible causes: cartridge stored above 35 °C, expired cartridge, or manufacturing defect (see **FC-FLD-03**). Discard the cartridge and retry with a new one from a different lot. Report recurring issues to MTS support with the lot number (printed on the foil pouch).

#### Q12: The cartridge is leaking after insertion.

A small amount of visible liquid at the cartridge inlet after sample application is normal (overflow channel). If liquid is leaking from the base or sides of the cartridge, this indicates a seal failure (**FC-FLD-02**) or housing crack (**FC-MEC-01**). Do not use the cartridge. Contact MTS support with the lot number and photos.

### 8.4 Ordering and Support

#### Q13: How do I order MDX-7000 cartridges?

Contact your regional OEM distributor or email [orders@medtech-solutions.de](mailto:orders@medtech-solutions.de). Minimum order quantity: 500 cartridges. Standard lead time: 4 weeks.

#### Q14: Is the MDX-7000 CE-marked?

Yes. The MDX-7000 is classified as an IVD device and carries the CE-IVDR mark under EU IVDR 2017/746. Technical documentation is available upon request under NDA.

## 9 Calibration Procedure — MDX-6100 Thermal Regulation Unit

### Calibration Procedure: MDX-6100 Thermal Regulation Unit

**Document ID:** DOC-QA-CP-009  
**Revision:** Rev 1.2  
**Effective Date:** 2025-10-18  
**Department:** Quality Assurance / Engineering

#### 9.1 Purpose

This procedure defines the calibration method for the MDX-6100 Thermal Regulation Unit, which provides precise temperature control for sample incubation chambers in clinical analyzers. Calibration ensures the unit maintains  $37.0 \pm 0.1\text{ }^{\circ}\text{C}$  in incubation mode as required by clinical assay protocols.

#### 9.2 Scope

Applicable to all MDX-6100 units during:

- Final production testing (PROC-QA-203)
- Periodic re-calibration (PROC-CAL-402)
- Post-repair verification

Calibration interval: **every 1,500 operating hours or 12 months**, whichever comes first.

#### 9.3 Reference Equipment

Table 13: Calibration Reference Equipment

| Item                          | ID           | Accuracy                                      | Cal. Due   |
|-------------------------------|--------------|-----------------------------------------------|------------|
| Precision thermometer (Pt100) | REF-THERM-01 | $\pm 0.02\text{ }^{\circ}\text{C}$            | 2026-08-15 |
| Data logger (8-channel)       | REF-LOG-03   | $\pm 0.01\text{ }^{\circ}\text{C}$ resolution | 2026-06-01 |
| Thermal paste (reference)     | REF-TP-01    | $\lambda = 8.5\text{ W/mK}$                   | N/A        |
| Ambient temperature monitor   | REF-AMB-02   | $\pm 0.1\text{ }^{\circ}\text{C}$             | 2026-09-20 |

#### 9.4 MDX-6100 Key Components

| Part Number | Description                                           | Role in Calibration         |
|-------------|-------------------------------------------------------|-----------------------------|
| PN-TR-6101  | Peltier element, 40 W                                 | Heating/cooling actuator    |
| PN-TR-6102  | NTC thermistor, 10 k $\Omega$ @ 25 $^{\circ}\text{C}$ | Temperature feedback sensor |
| PN-TR-6103  | Aluminum heat spreader plate                          | Thermal uniformity          |

#### 9.5 Calibration Procedure

**Environment:** Ambient temperature  $22 \pm 3\text{ }^{\circ}\text{C}$ , humidity 30–60% RH. Allow all equipment to thermally stabilize for 30 minutes before starting.

##### 1. Setup.

- a) Place the MDX-6100 on a non-conductive surface.
- b) Apply thermal paste (REF-TP-01) to the Pt100 probe tip.
- c) Insert REF-THERM-01 into the calibration port (marked “CAL” on the unit housing).
- d) Connect REF-LOG-03 to the Pt100 output and the MDX-6100 NTC output (test points TP1 and TP2 on the control PCB).
- e) Connect the MDX-6100 to the 24 VDC power supply (MTS-PSU-24V-5A).

**2. Ambient Baseline.** Record ambient temperature from REF-AMB-02. If outside 19–25 °C, do not proceed; adjust room temperature or relocate.

**3. Setpoint Test — 37.0 °C (Incubation Mode).**

- a) Send command `CMD_SET_TEMP 37.0` via serial interface (115200 baud, 8N1).
- b) Wait until the MDX-6100 reports “STABLE” status (typically 90–120 s).
- c) Record temperature from REF-THERM-01 every 10 s for 5 minutes (30 data points).
- d) **Pass criteria:** All readings within  $37.0 \pm 0.1$  °C.
- e) If readings are outside tolerance, proceed to Step 7 (Adjustment).

**4. Setpoint Test — 25.0 °C (Storage Mode).**

- a) Send `CMD_SET_TEMP 25.0`.
- b) Wait for “STABLE” (typically 60–90 s).
- c) Record for 5 minutes.
- d) **Pass criteria:** All readings within  $25.0 \pm 0.2$  °C.

**5. Ramp Rate Test.**

- a) From 25.0 °C, send `CMD_SET_TEMP 37.0`.
- b) Measure time to reach 36.9 °C.
- c) **Pass criteria:** Ramp time  $\leq 120$  s. Overshoot  $\leq 0.5$  °C (i.e., temperature must not exceed 37.5 °C during ramp).
- d) If overshoot  $> 0.5$  °C, record as **FC-THR-02**.

**6. Uniformity Test.**

- a) At 37.0 °C stable, measure temperature at 5 points across the heat spreader plate (PN-TR-6103): center and four corners.
- b) **Pass criteria:** Maximum deviation between any two points  $\leq 0.15$  °C.

**7. Adjustment (if required).**

- a) If 37.0 °C test fails, adjust the NTC offset via `CMD_CAL_NTC_OFFSET <value>` (range:  $-2.0$  to  $+2.0$  °C, resolution 0.01 °C).
- b) Repeat Step 3. If still out of spec, check Peltier element COP: send `CMD_PELTIER_DIAG`. If  $COP < 0.8$ , replace PN-TR-6101 and re-calibrate. Log **FC-THR-01**.
- c) If NTC drift exceeds 0.2 °C from the previous calibration, replace PN-TR-6102. Log **FC-THR-03**.

**8. Documentation.**

- a) Record all data in the Calibration Certificate (template MTS-CAL-CERT-6100).
- b) Affix calibration label to the unit with: date, due date, technician initials.
- c) Upload data to MTS-MES-Pro under the unit serial number.
- d) If any failure code was logged, initiate deviation report per SOP-QA-DEV-01.



## 9.6 Acceptance Criteria Summary

Table 14: MDX-6100 Calibration Acceptance Criteria

| Test                 | Criterion                           | Failure Code (if out of spec) |
|----------------------|-------------------------------------|-------------------------------|
| 37.0 °C stability    | $\pm 0.1\text{ }^{\circ}\text{C}$   | FC-THR-03 (if NTC drift)      |
| 25.0 °C stability    | $\pm 0.2\text{ }^{\circ}\text{C}$   | FC-THR-03                     |
| Ramp time (25→37 °C) | $\leq 120\text{ s}$                 | —                             |
| Ramp overshoot       | $\leq 0.5\text{ }^{\circ}\text{C}$  | FC-THR-02                     |
| Uniformity           | $\leq 0.15\text{ }^{\circ}\text{C}$ | —                             |
| Peltier COP          | $\geq 0.8$                          | FC-THR-01                     |

## 10 Incident Report — IR-2026-017

### Incident Report: IR-2026-017

**Document ID:** DOC-QA-IR-010  
**Revision:** Rev 1.0  
**Effective Date:** 2026-03-18  
**Department:** Quality Assurance

#### 10.1 Incident Summary

| Field             | Value                                                           |
|-------------------|-----------------------------------------------------------------|
| Incident ID       | IR-2026-017                                                     |
| Date of discovery | 2026-03-12                                                      |
| Date of report    | 2026-03-18                                                      |
| Reported by       | Jens Krickl, Production Manager                                 |
| Product affected  | MDX-4200-R3 Optical Sensor Array Module                         |
| Batch affected    | PO-2026-0312 (60 units)                                         |
| Severity          | Medium (no patient safety impact; product performance degraded) |
| Status            | Open — Under Investigation                                      |

#### 10.2 Description of Incident

During routine final functional testing ([PROC-QA-203](#)) on 2026-03-12, the QA technician observed that 12 out of 60 units in batch PO-2026-0312 of the MDX-4200-R3 exhibited elevated channel-to-channel crosstalk values. Specifically, adjacent-channel crosstalk was measured at  $-38$  to  $-42$  dB, exceeding the specification limit of  $< -45$  dB (see DOC-ENG-DS-001, Section 1).

The affected units passed all other functional tests (dark current, LED intensity, ADC linearity, filter wheel indexing). No failure code was automatically triggered because the crosstalk parameter was checked manually as part of the extended test protocol introduced in Rev 3.2 of the test specification (TS-MDX4200-R3-v3.2).

The 12 affected units were quarantined under non-conformance report NCR-2026-0045.

#### 10.3 Immediate Containment Actions

1. All 60 units of batch PO-2026-0312 placed on hold pending investigation (2026-03-12, J. Krickl).
2. Remaining 48 units re-tested for crosstalk: 45 passed, 3 additional failures identified. Total affected: 15/60 units (25%).
3. Production of MDX-4200-R3 suspended for batches using alignment fixture EQ-CR12-004 pending root cause determination (2026-03-13, M. Brenner).
4. Customer shipments of MDX-4200-R3 from stock reviewed: no affected units had been shipped (batch was in WIP status).

#### 10.4 Root Cause Investigation

### 10.4.1 Timeline

| Date       | Activity                                                                                  |
|------------|-------------------------------------------------------------------------------------------|
| 2026-03-12 | Failure detected in <a href="#">PROC-QA-203</a> . Units quarantined.                      |
| 2026-03-13 | Engineering review of assembly records in MES.                                            |
| 2026-03-14 | Optical alignment data retrieved for batch PO-2026-0312.                                  |
| 2026-03-15 | Alignment fixture EQ-CR12-004 found to have laser drift (see DOC-PROD-ML-006, Section 6). |
| 2026-03-17 | Laser module replaced; fixture re-calibrated.                                             |

### 10.4.2 Root Cause

The alignment laser on fixture EQ-CR12-004 experienced thermal drift due to prolonged operation during a high-volume production run (3 consecutive shifts on 2026-03-10 to 2026-03-12). The laser mounting bracket, made of aluminum, expanded by approximately  $8\text{ }\mu\text{m}$  over the 18-hour continuous run, causing a systematic offset in the collimating lens (PN-0S-4202) alignment for channels 7–12.

This offset resulted in increased optical crosstalk between adjacent channels in the affected region. The root cause was confirmed by correlating the MES-recorded alignment positions (Step 5 of DOC-PROD-SOP-002, Section 2) with the time-of-assembly for each unit. Units assembled after hour 10 of the continuous run showed progressively worse crosstalk values.

### 10.4.3 Contributing Factors

- The alignment fixture’s preventive maintenance schedule ([PROC-QA-202](#)) did not include a thermal stability check during extended production runs.
- The in-process optical verification ([PROC-QA-202](#)) checks only centroid deviation ( $< 5\text{ }\mu\text{m}$ ), not crosstalk directly. The  $8\text{ }\mu\text{m}$  drift was marginal and did not trigger the centroid alarm.
- No automated thermal compensation was implemented on EQ-CR12-004.

## 10.5 Corrective and Preventive Actions

Table 15: CAPA Actions for IR-2026-017

| CAPA ID | Action                                                                                                                 | Owner           | Due Date   | Status      |
|---------|------------------------------------------------------------------------------------------------------------------------|-----------------|------------|-------------|
| C-017-1 | Install thermal insulation pad on EQ-CR12-004 laser bracket                                                            | Dr. L. Pfeiffer | 2026-03-20 | Completed   |
| C-017-2 | Add mandatory 30-min cool-down after 8h continuous operation to SOP DOC-PROD-SOP-002                                   | J. Krickl       | 2026-03-25 | In progress |
| C-017-3 | Tighten in-process centroid alarm from $5\text{ }\mu\text{m}$ to $3\text{ }\mu\text{m}$ in <a href="#">PROC-QA-202</a> | Dr. S. Hartmann | 2026-04-01 | Planned     |
| C-017-4 | Evaluate active thermal compensation (Peltier-based) for fixture EQ-CR12-004                                           | M. Brenner      | 2026-05-15 | Planned     |
| C-017-5 | Add crosstalk measurement to automated final test sequence (currently manual)                                          | Dr. L. Pfeiffer | 2026-04-30 | Planned     |

## 10.6 Disposition of Affected Units

- **15 units (crosstalk out of spec):** Reworked. Collimating lens PN-0S-4202 re-aligned using the repaired fixture. All 15 units re-tested and passed ( $< -45$  dB). Released to stock on 2026-03-22.
- **45 units (passed re-test):** Released to stock on 2026-03-15.

## 10.7 Regulatory Assessment

This incident does not constitute a reportable event under EU IVDR 2017/746 Article 87 because:

- No affected units were distributed to customers.
- The failure mode (elevated crosstalk) would cause reduced measurement accuracy but not a safety hazard.
- All units were contained within the MTS facility.

The incident is documented in the post-market surveillance technical file for the MDX-4200-R3 product family.

## 10.8 Approval

| Role                | Name                | Date       |
|---------------------|---------------------|------------|
| QA Director         | Dr. Simone Hartmann | 2026-03-18 |
| Production Manager  | Jens Krickl         | 2026-03-18 |
| Head of Engineering | Dr. Lukas Pfeiffer  | 2026-03-18 |
| CTO                 | Markus Brenner      | 2026-03-19 |

## Document Index

Table 16: \*

| Complete Document Index — MedTech Solutions GmbH RAG Knowledge Base |                                |                 |               |
|---------------------------------------------------------------------|--------------------------------|-----------------|---------------|
| Doc ID                                                              | Title                          | Type            | Dept.         |
| DOC-ENG-DS-001                                                      | MDX-4200-R3 Datasheet          | Datasheet       | Engineering   |
| DOC-PROD-SOP-002                                                    | Cleanroom Assembly SOP         | SOP             | Production    |
| DOC-CS-TSG-003                                                      | MDX-4200 Troubleshooting Guide | TSG             | Cust. Support |
| DOC-QA-QR-004                                                       | Q1 2026 Quality Report         | Quality Report  | QA            |
| DOC-ENG-SS-005                                                      | Supplier Spec (SUP-001)        | Supplier Spec   | Engineering   |
| DOC-PROD-ML-006                                                     | Maintenance Log CR-A12         | Maint. Log      | Production    |
| DOC-QA-CC-007                                                       | ISO 13485 Compliance Checklist | Checklist       | QA            |
| DOC-CS-FAQ-008                                                      | MDX-7000 Customer FAQ          | FAQ             | Cust. Support |
| DOC-QA-CP-009                                                       | MDX-6100 Calibration Procedure | Cal. Procedure  | QA            |
| DOC-QA-IR-010                                                       | Incident Report IR-2026-017    | Incident Report | QA            |

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